

ME557- Project 2: Sphere of Petals

Design Description and Experience:

The Sphere of Petals is a $4 \times 4 \times 4 \text{ cm}^3$ design with seven circular cutouts aligned to create a symmetrical flower pattern on a hollow sphere. The top and the bottom surfaces both mirrored this flower motif, creating an eye-catching, three-dimensional look. To construct this design, the sphere was modelled in CAD, with the circular forms extruded to produce the flower pattern.

Initially, a $2 \times 2 \times 2 \text{ cm}^3$ scale with a wall thickness of 1 mm was used to test the original concept. A FormLabs SLA printer with clear resin was used to print these smaller spheres twice. However, they were incredibly delicate. The fragile construction would break whenever external supports were removed with snippers, illustrating the constraints of printing extremely thin and microscopic geometries.

The design was scaled up to $4 \times 4 \times 4 \text{ cm}^3$ with a 2 mm increase in the sphere thickness in order to overcome these difficulties. Because of the adequate structural strength this modification offered, the external supports could be removed without causing any harm. The final design, however, retained some internal supports that could not be removed due to its intricacy of the hollow interior. In spite of this, the sphere managed to maintain its hollow geometry and complex floral pattern, which would be difficult to produce using conventional subtractive methods.

This project demonstrated the creative possibilities of SLA 3D printing as well as the iterative design process needed to maximize printability, strength, and aesthetics. The finished Sphere of Petals demonstrates the benefits of additive manufacturing for complicated interior elements and delicate patterns that conventional machines cannot accomplish by combining aesthetic appeal, structural durability, and geometric intricacy.



Fig 1. Picture of the Sphere of Petals printed with an SLA printer.